

Graphormer: Reproduction, Ablation, and Inference-Time Architectural Extensions

CS768 – Course Project, Endsem Report

Harshil Amitkumar Solanki (23B1016)

Sumedh S S (23B1079)

Abstract

We re-implement Graphormer [1] from scratch as `mini_graphormer`, a 48M-parameter PyTorch package free of the fairseq and Cython dependencies that block the original release. Microsoft’s `pcqm4mv2_graphormer_base` checkpoint loads with *strict* parameter matching, and on a 2,000-graph slice of the PCQM4Mv2 validation set our reproduction reaches $\text{MAE} = 0.102 \text{ eV}$ – consistent with the published 0.0864 eV within finite-sample variance. An ablation of the four structural biases reveals that the multi-hop *edge* bias dominates (+338% MAE when removed), far ahead of the shortest-path *spatial* bias (+36%), contradicting the paper’s framing. We then test three inference-time architectural extensions inspired by SAN, GTN, and PAGTN – Laplacian PE, RWPE, and a ring-aware bias. All three converge to a null result: the trained model is locked to its specific bias signature and cannot integrate post-hoc structural priors.

1 Implementation

The original Graphormer release ships as a fairseq fork with a Cython preprocessing module; both have well-known installation issues on modern toolchains. We re-implement the model and pipeline as a self-contained PyTorch package. Every parameter is named identically to the released checkpoint, so `model.load_state_dict(state_dict, strict=True)` succeeds with *zero* missing and *zero* unexpected keys (all 210 tensors aligned, including embeddings, attention projections, and the regression head). The PCQM4Mv2 data path replaces the Cython all-pairs Floyd–Warshall and multi-hop edge encoder with vectorised NumPy, and streams the validation split directly from the raw CSV, sidestepping OGB’s full-dataset upfront SMILES processing (3.7M molecules, hours on CPU). Both ablations and extensions are implemented as forward hooks on `nn.Modules`; underlying weights are never modified, allowing many configurations to be evaluated against a single loaded checkpoint.

2 Reproducing the pretrained baseline

On the first 2,000 PCQM4Mv2 validation graphs (CSV-ordered prefix, no shuffling), our implementation reaches **$\text{MAE} = 0.1020 \text{ eV}$** , against the published **0.0864 eV** on the full 73,545-graph set. We verified separately ($n = 1000$ sweep) that within-slice MAE swings by roughly $\pm 0.01 \text{ eV}$ depending on which contiguous 1000 graphs are sampled; the 0.016 eV gap to the published number is therefore consistent with sampling noise on a non-shuffled prefix, not with any architectural defect (which the strict-load test independently rules out).

3 Ablation study

Each ablation zeroes the output of one structural-bias submodule via a forward hook, leaving the rest of the model untouched. Results (Table 1) show a striking and *counter-intuitive* hierarchy.

Table 1: Effect of removing each Graphormer-specific structural bias on PCQM4Mv2 validation MAE ($n = 2000$).

removed bias	MAE	$\Delta \text{ MAE}$	$\Delta \%$
baseline (full model)	0.1020	—	—
spatial bias $b(\phi(i, j))$	0.1384	+0.036	+36%
multi-hop edge bias $c(i, j)$	0.4471	+0.345	+338%
virtual graph-token distance	0.1020	+0.000	+0%
in/out-degree centrality	0.1178	+0.016	+16%
all attention biases together	0.7576	+0.656	+643%

Quantitative findings. (i) The multi-hop edge bias is by far the most load-bearing component: removing it alone quadruples MAE, roughly nine times the impact of removing the spatial bias the original paper foregrounds. (ii) The graph-token virtual-distance term has zero measurable effect on validation MAE – it can be ablated freely without penalty. (iii) Centrality (degree encoding on node features) contributes modestly (+16%). (iv) The sum of individual ablation costs (390%) falls well short of the all-bias-removed cost (643%), indicating super-additive interactions between biases beyond the linear regime.

Qualitative reading. Graphormer is usually summarised as “Transformer + spatial encoding,” yet on PCQM4Mv2 the multi-hop edge encoding is what actually carries the structural prior. The model’s predictions

degenerate to a near-mean predictor when all attention biases are removed ($\text{MAE}=0.76$), confirming that the transformer backbone alone – with no graph-aware bias – recovers little beyond label statistics on this benchmark.

4 Inference-time architectural extensions

Our midsem literature review covered three relatives of Graphormer that each enrich its structural prior in a different way: SAN (spectral) [2], GTN (random-walk-style adjacency combinations) [3], and PAGTN (path / ring features on molecular graphs) [4]. We extract the canonical structural signal from each and add it as an *additive attention bias*, on top of Graphormer’s existing (b, c, t) terms, with one tunable scalar α :

- **Laplacian PE** (SAN-flavoured): $\alpha U_k W U_k^\top$, where U_k are the smallest non-trivial Laplacian eigenvectors per molecule.
- **RWPE** (GTN-flavoured): $\alpha \sum_{k=1}^K (D^{-1/2} A D^{-1/2})^k$, a multi-scale soft adjacency built from random-walk powers.
- **Ring-aware** (PAGTN-flavoured): $\alpha \cdot \mathbf{1}[i, j \text{ share a ring}]$, with ring identity extracted via rdkit’s SSSR.

For each extension we sweep α on the same $n=2000$ slice; the best “all + extension” configuration (full Graphormer plus the new bias) is reported in Table 2.

Table 2: Best inference-time extension per method after sweeping α on the $n=2000$ slice. The “X only” column is for context: with all of Graphormer’s structural biases zeroed, the new term alone cannot replace them and MAE collapses by 6–7 $\times$.

extension	best α	MAE (all + ext)	Δ vs base	MAE (ext only)
baseline	—	0.10197	—	—
Laplacian PE	0.5	0.10182	−0.00015	0.7285
RWPE	0.05–0.1	0.10190	−0.00007	0.7096
Ring-aware	0.05	0.10192	−0.00005	0.7041

All three extensions land on a *sub-millivolt* improvement at very small α , with monotonic degradation as α grows: e.g. RWPE $\alpha=1.8$ inflates MAE by +8%; ring $\alpha=2$ by +57%. The best gain (Laplacian, −0.00015 eV) is roughly 1/30 of the within-slice MAE noise floor – statistically indistinguishable from zero.

Single-molecule sanity check. On five canonical aromatic systems (Table 3) the differences between extensions are ≤ 0.1 eV. The cleanest signal is on *pyrene*, where the ring-aware bias shifts the prediction 0.08 eV toward the experimental gap – a small but structurally-justified move, consistent with the a-priori prediction that 4-fused-ring topology benefits from explicit same-ring information that distance alone cannot encode.

Table 3: Per-molecule HOMO–LUMO gap predictions in eV. Experimental B3LYP gaps from literature. Extension hyperparameters at default values ($\alpha_{\text{Lap}}=1.0$, $\alpha_{\text{RWPE}}=0.5$, $\alpha_{\text{ring}}=1.0$).

molecule	exp.	baseline	+Lap.	+RWPE	+Ring
naphthalene	4.40	4.834	4.830	4.834	4.840
azulene	1.70	3.333	3.336	3.308	3.223
anthracene	3.30	3.586	3.588	3.579	3.580
phenanthrene	3.90	4.748	4.752	4.750	4.742
pyrene	3.30	3.842	3.843	3.834	3.760

5 Discussion

Why the null result is informative. Three structurally distinct priors – spectral, random-walk, ring-binary – all converge to roughly the same near-zero effect. The bottleneck is therefore not the *kind* of structural information added but the model’s inability to integrate *any* new bias post-hoc. The 12 transformer layers were trained to a specific scale and shape of Graphormer’s $(b(\phi), c(i, j), t)$ signature; perturbing the input distribution to softmax with even modest additional bias trades calibration loss for whatever new signal the prior carries, and the trade-off is roughly break-even. We expect that the same priors, applied during training (or even brief fine-tuning), would behave very differently – the model could re-calibrate its bias scale to absorb the new term.

The edge-bias finding deserves a follow-up. That removing $c(i, j)$ alone quadruples MAE – four times the impact of removing the spatial bias the original paper foregrounds – suggests the multi-hop edge encoder is the architecture’s actual workhorse on molecular regression. A natural training-time direction, motivated by PAGTN [4], is to enrich edge features along shortest paths (path-level node-pair features, ring-membership of intermediate atoms) rather than further refining node-pair distance, where Graphormer is already saturated.

Limitations. Our extensions are tested in a frozen-weight setting; the picture would likely change with even brief fine-tuning. The slice size $n=2000$ imposes a noise floor of $\approx 0.003\text{eV}$ that any inference-time extension would need to clear to be considered a real gain. The handcrafted per-molecule predictions are illustrative, not ground truth – common PAHs are likely *in* the PCQM4Mv2 training set, which means the baseline already has access to memorised priors that no inference-time perturbation can dislodge. Code is available at <https://github.com/HARSHIL2836R/Graphormer-Project>.

References

- [1] C. Ying, T. Cai, S. Luo, S. Zheng, G. Ke, D. He, Y. Shen, T.-Y. Liu. Do transformers really perform bad for graph representation? *NeurIPS*, 2021.
- [2] D. Kreuzer, D. Beaini, W. Hamilton, T. Leman, M. Bronstein. Rethinking graph transformers with spectral attention. *NeurIPS*, 2021.
- [3] S. Yun, M. Jeong, R. Kim, J. Kang, H. J. Kim. Graph transformer networks. *NeurIPS*, 2019.
- [4] B. Chen, R. Barzilay, T. Jaakkola. Path-augmented graph transformer network. *ICML Workshop on Learning and Reasoning with Graph-Structured Data*, 2019.